Metallic Bonding and Types of Solids

Answer key with explanations

A. Metallic bonding

1. Metallic bonding.

> (a) The attraction between **positive metal ions** in a regular lattice and a **sea of delocalized electrons**. (b) The delocalized electrons are **free to move**, so they carry a current. The layers of ions can **slide** over each other while the electron sea keeps holding them together, so the metal changes shape without breaking.

2. Magnesium and sodium.

> Magnesium gives **two** electrons per atom to the sea (forming Mg^{2+}), sodium only one (Na^{+}). Magnesium ions are also **smaller** and more highly charged, so the attraction between the ions and the electron sea is **stronger**, and more energy is needed to melt it.

3. Alloys.

> (a) A mixture of a metal with other elements (usually metals).
>
> (b) Tin atoms are a different size from copper atoms, so they **disrupt the layers**, which can no longer slide easily: the alloy is harder.
>
> (c) $m_{\mathrm{Cu}} = \dfrac{88.0}{100} \times 40.0\ \mathrm{g} = \mathbf{35.2\ g}$ and $m_{\mathrm{Sn}} = 40.0\ \mathrm{g} - 35.2\ \mathrm{g} = \mathbf{4.8\ g}$

B. Types of solid

4. Classify.

> P: **metallic** (conducts as a solid: mobile electrons). Q: **molecular** (low melting point, no conduction). R: **ionic** (conducts only when the ions are free to move). S: **covalent network** (strong covalent bonds throughout, no mobile charges).

5. Explanations.

> (a) In solid KCl the ions are fixed in the lattice. In solution they are **free to move** and carry charge. (b) Iodine is a **molecular** solid. Melting only overcomes the weak **forces between** I_2 molecules; the I–I covalent bonds are not broken.

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C. Diamond and graphite

6. Diamond and graphite.

> (a) In diamond every carbon is covalently bonded to **four** others in a rigid 3D network; many strong bonds must be broken to scratch or break it. (b) In graphite each carbon bonds to **three** others in flat layers; the fourth outer electron is **delocalized** along the layer and can move, carrying a current. (c) The layers are held together only by **weak forces**, so they slide off onto the paper.

D. Comparing structures

7. Bend or shatter.

> In the metal, layers of positive ions slide over each other but stay surrounded by the electron sea, so the bonding is not broken: it **bends**. In salt, a slight shift brings ions of the **same charge** next to each other; they repel and the crystal **shatters**.

E. Challenge

8. SiO_2 and CO_2.

> CO_2 is **molecular**: separate small molecules held together only by weak forces, which are overcome far below room temperature. SiO_2 is a **covalent network**: each Si is bonded to four O atoms and each O to two Si atoms in a giant 3D structure, so melting requires breaking many strong **covalent bonds**. The formula SiO_2 gives only the ratio of atoms, not a molecule.

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